CHAPTER 13

State Machines (FSM)

State · state diagram · state table · Moore and Mealy · state reduction · sequence detector

QUICK TRICK
Two lines to remember: Moore = output is written **inside the circle** (depends on the state only).
Mealy = output is written **on the arrow** as input/output (depends on state and input).

13.1 What is a state machine?

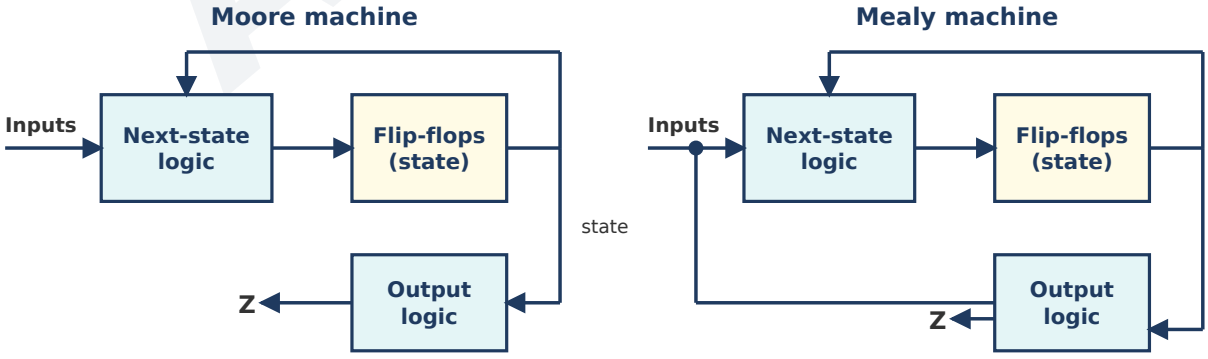
● **LEVEL 1 · BEGINNER**

What is it? A **finite state machine (FSM)** is a sequential circuit that is in **one state at a time**. When an input arrives (at a clock pulse) it moves to the next state, and it may give an output.

Why do we need it? To design anything that follows steps: counters, traffic lights, vending machines, serial data checkers.

Analogy. A lift. Its states are “at floor 1”, “at floor 2”… A button press (input) moves it to a new state.

Word	Simple meaning
State	A memory of what has happened so far (for example “I have seen a 1”).
Input / Output	What goes into the machine / what comes out.
Transition	A move from one state to another.
State diagram	A picture: circles = states, arrows = transitions.
State table	The same information written as a table.
Present state / Next state	Where we are now / where we go after the clock edge.



In a **Moore** machine the output logic looks only at the state. In a **Mealy** machine the output logic looks at the state **and** the input.

13.2 Moore vs Mealy

● LEVEL 2 · EXAM

	Moore machine	Mealy machine
Output depends on	**Present state only**	**Present state AND input**
In the diagram	Output **inside** the circle (S1/0)	Output **on the arrow** (input/output, like 1/0)
Number of states	Usually **more**	Usually **fewer**
When the output changes	Only after a clock edge (when the state changes)	Can change as soon as the input changes
Output glitches	Safer (stable between clock edges)	Can glitch if the input is noisy
Speed of reaction	One clock cycle later	Reacts in the same cycle

QUICK TRICK

Moore → Mealy: put the output of the **destination state** on every arrow that enters it. The number of states does not grow (it can only stay the same or become smaller).
Mealy → Moore: make one state for each pair (state, output) that appears on the arrows entering it. The number of states can grow.

13.3 Design steps

● LEVEL 2 · EXAM

```
Word problem
     |
     v
STATE DIAGRAM  ->  STATE TABLE  ->  STATE REDUCTION (merge equal states)
     |
     v
STATE ASSIGNMENT (give each state a binary code)
     |
     v
Pick flip-flops (D is easiest)  ->  Write equations (K-maps)  ->  Draw the circuit
```

QUICK TRICK

Number of flip-flops = $\lceil \log_2 N \rceil$ for N states. 2 states → 1 FF. 3–4 states → 2 FFs. 5–8 states → 3 FFs. 9–16 states → 4 FFs.

QUICK TRICK

Unused codes = don't cares. With 3 states you need 2 FFs, so code 11 is unused. Treat its outputs and next states as X in the K-maps. This often makes the logic smaller.

QUICK TRICK

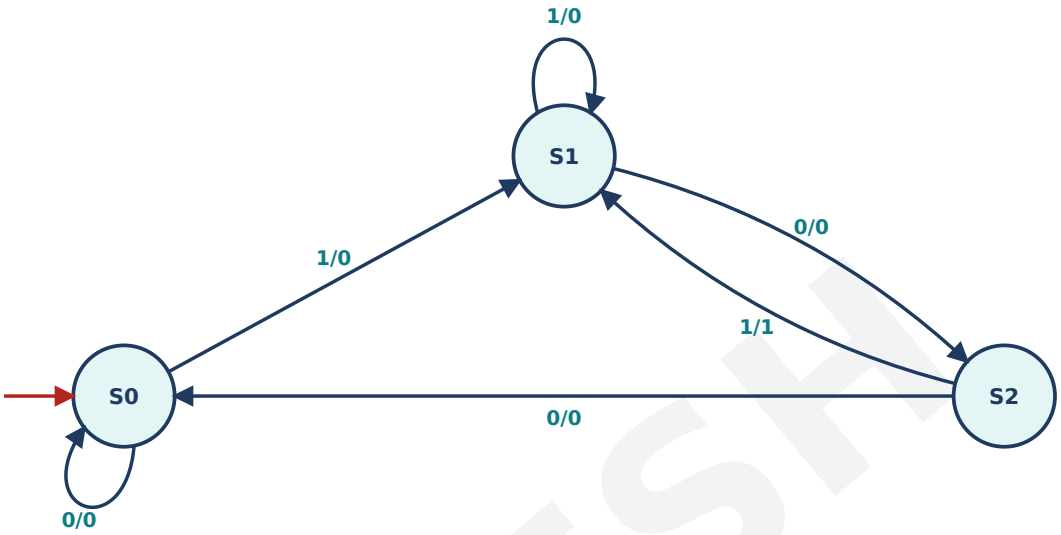
Every state needs an arrow for every input value. With 1 input bit, each state has 2 arrows out (for 0 and for 1). With I input bits, each state has 2^I arrows. The state table has N rows and 2^I columns.

13.4 Worked example: sequence detector “101” (Mealy)

● LEVEL 2 · EXAM

Problem. Bits arrive one by one on input x. The output Z must be 1 **when the last three bits received are 1, 0, 1**. Overlapping is allowed (the last 1 of one 101 can start the next one).

States (what we remember): S0 = nothing useful yet. S1 = the last bit was 1. S2 = the last two bits were 1 0.



Arrow labels are **input / output** (Mealy). The only arrow with output 1 is S2 → S1 on input 1: we had seen “10” and now got a 1.

Present state	x = 0: Next state / Z	x = 1: Next state / Z
S0	S0 / 0	S1 / 0
S1	S2 / 0	S1 / 0
S2	S0 / 0	S1 / 1

Trace. Input 1 0 1 0 1 1, start in S0:

Input x	1	0	1	0	1	1
Next state	S1	S2	S1	S2	S1	S1
Output Z	0	0	1	0	1	0

QUICK TRICK

To trace any machine: write the present state, read the arrow for the input, write the next state and the output. Repeat for every input bit.

Building the circuit (D flip-flops)

3 states need 2 flip-flops (Q1, Q0). Choose the codes: **S0 = 00, S1 = 01, S2 = 10**. The code 11 is unused (X).

Q1	Q0	x	D1 (Q1+)	D0 (Q0+)	Z
0	0	0	0	0	0
0	0	1	0	1	0
0	1	0	1	0	0

Q1	Q0	x	D1 (Q1+)	D0 (Q0+)	Z
0	1	1	0	1	0
1	0	0	0	0	0
1	0	1	0	1	1
1	1	0	X	X	X
1	1	1	X	X	X

Using K-maps (the X rows help us make big groups) we get very simple equations:

```
D1 = Q0 . x'        (next Q1 = 1 only when we are in S1 and the input is 0)
D0 = x              (next Q0 = the present input)
Z  = Q1 . x         (output 1 only in S2 with input 1)
```

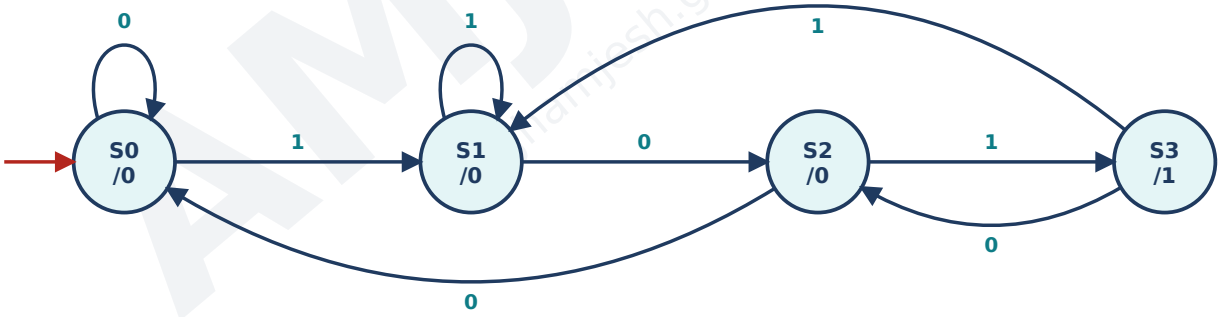
QUICK TRICK

Check: start in 00 and feed 1,0,1,1,0,1. The circuit gives Z = 0,0,1,0,0,1. The pattern 101 appears at the 1st–3rd and the 4th–6th bits, so Z = 1 at the 3rd and 6th bit. ✔

13.5 The same detector as a Moore machine

● **LEVEL 2 · EXAM**

In a Moore machine the output belongs to the state, so we need an extra state S3 whose output is 1 (“I have just seen 101”).



Present state	Output Z	x = 0: Next	x = 1: Next
S0	0	S0	S1
S1	0	S2	S1
S2	0	S0	S3
S3	1	S2	S1

GATE POINT

Counting rule for a sequence detector of length k: Mealy needs **k** states, Moore needs **k + 1** states. For “101”: Mealy 3 states (2 FFs), Moore 4 states (2 FFs). For “1011”: Mealy 4 states (2 FFs), Moore 5 states (3 FFs).

EXAM TRAP

In the Moore diagram the output is written inside the circle (S3/1). If you write it on the arrow you have drawn a Mealy machine.

13.6 State reduction

● LEVEL 3 · GATE

Why? Fewer states means fewer flip-flops and fewer gates. **Two states are equivalent** if, for every input, they give the **same output** and go to the **same (or equivalent) next states**. Equivalent states can be merged.

Example (Mealy, next state / output):

Present	x = 0	x = 1
A	A / 0	B / 0
B	A / 0	C / 1
C	A / 0	D / 1
D	A / 0	D / 1

```
STEP 1  Group states with the same OUTPUTS:   {A}   {B, C, D}
        (B, C, D all give 0 on x = 0 and 1 on x = 1)
STEP 2  Check the next states of {B, C, D}:
           x = 0 : all go to A                  (same group)
           x = 1 : B -> C, C -> D, D -> D       (all inside {B, C, D})
        Nothing splits, so B, C and D are EQUIVALENT.
RESULT  4 states -> 2 states.  Number of flip-flops: 2 -> 1.
```

QUICK TRICK

Merge-identical-rows trick: look for two rows in the state table that are exactly the same (same outputs, same next states). Merge them, rename, and look again. Here the rows of C and D are almost the same: on x = 1, C goes to D and D goes to D. So C ≡ D, and we merge C into D. Now B (x = 1 → D/1) has exactly the same row as D, so B ≡ D too.

Reduced machine: A: x = 0 → A/0, x = 1 → B'/0. B': x = 0 → A/0, x = 1 → B'/1. (It outputs 1 when the previous input and the present input are both 1.)

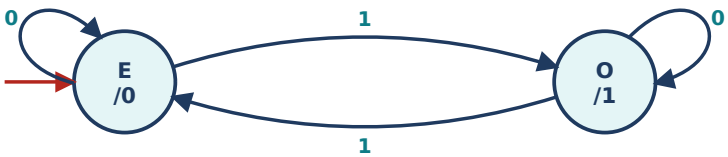
COMMON MISTAKE

Merging states that have the same next states but **different outputs**. Outputs must match first.

13.7 A small Moore example: odd-parity detector

● LEVEL 2 · EXAM

Problem. Output Z = 1 if the number of 1s received so far is **odd**. **States:** E (even number so far, output 0), O (odd number, output 1).



One flip-flop is enough (2 states). Let Q = 0 for E and Q = 1 for O. The state toggles when x = 1 and stays when x = 0. That is exactly a **T flip-flop with T = x**, and the output is **Z = Q**. (Or a D

flip-flop with D = Q ⊕ x.)

> **QUICK TRICK**
> **Pattern:** "change state on input 1, stay on input 0" = T flip-flop with T = x. You already know this from Chapters 11 and 12.

Quick-Solve Tricks — Summary

Topic	Trick
Moore or Mealy?	Output inside the circle → Moore. Output on the arrow (x/z) → Mealy.
Flip-flops needed	$\lceil \log_2 N \rceil$.
Sequence detector states	Pattern length k → Mealy k, Moore k + 1.
Tracing	Present state + input → next state + output. Repeat.
State reduction	Same outputs first, then same next states. Merge identical rows, repeat.
Unused codes	Use as don't cares (X).
Moore → Mealy	Put the destination state's output on the arrows entering it.
Mealy → Moore	One state per (state, output) pair. States can increase.
Table size	N rows × 2^I input columns.
Easiest flip-flop to use	D flip-flop: D = next state.

GATE QUICK CHECK — Chapter 13

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Moore : Z = f(state)                Mealy : Z = f(state, input)
Flip-flops = ceil(log2 N)
Detector of length k : Mealy k states, Moore k+1 states
State table : N rows, 2^I columns
Equivalent states : same outputs and equivalent next states
```

Traps: output position in the diagram · Mealy needs fewer states · Moore output is one clock later · unused state codes are don't cares · merge only states with the same outputs.

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. On what does the output of a Mealy machine depend?

Q2. How many flip-flops are needed for a machine with 9 states?

Q3. Where is the output written in a Moore state diagram?

● **LEVEL 2 · EXAM**

Q4. For the Mealy "101" detector of this chapter, find the outputs for the input 1 0 1 0 1 1.

Q5. How many states does a Moore "1101" detector need? How many does the Mealy version need?

Q6. A Mealy machine has states A, B, C. A: x=0 → B/0, x=1 → A/1. B: x=0 → A/0, x=1 → C/1. C: x=0 → A/0, x=1 → C/1. Can any states be merged?

● **LEVEL 3 · GATE**

Q7. Using the circuit D1 = Q0·x', D0 = x, Z = Q1·x (start 00), find Z for the inputs 1, 0, 1, 1, 0, 1.

Q8. A machine has 5 states and 2 input bits. How many entries are in its next-state table? How many flip-flops are needed?

Q9. Which type of machine can respond to an input change in the same clock cycle?

Q10. A Mealy machine has two states: A (previous bit 0) and B (previous bit 1). It outputs 1 only when it is in B and x = 1. How many states does an equivalent Moore machine need?

Q11. How many flip-flops does the Mealy "101" detector need? Why is the code 11 a don't care?

Q12. Why is the output of a Moore machine free of glitches caused by input noise between clock edges?

Answers

Q	Solution
1	**The present state and the present input.**
2	$\lceil \log_2 9 \rceil$ = **4** flip-flops ($2^3 = 8$ is not enough).
3	**Inside the circle**, next to the state name (like S3/1).
4	S0 → 1 → S1/0. S1 → 0 → S2/0. S2 → 1 → S1/1. S1 → 0 → S2/0. S2 → 1 → S1/1. S1 → 1 → S1/0. Outputs: **0 0 1 0 1 0**.
5	Pattern length k = 4. Moore: k + 1 = **5** states. Mealy: k = **4** states.
6	All three states give output 0 when x = 0 and output 1 when x = 1, so they start in one group {A, B, C}. Their next states: x = 0 → B, A, A and x = 1 → A, C, C. All of these lie inside the same group, so nothing splits. **A, B and C are all equivalent**: the machine reduces to **one state** with Z = x.
7	Trace: 00 → 01 → 10 → 01 → 01 → 10 → 01. Outputs **0, 0, 1, 0, 0, 1**.
8	5 states × 2^2 = 20 entries. $\lceil \log_2 5 \rceil$ = **3** flip-flops.
9	**Mealy** machine (the output depends directly on the input).
10	Moore states = (state, output) pairs on entering arrows: A (output 0), B with output 0, B with output 1. So **3 states**.
11	3 states → $\lceil \log_2 3 \rceil$ = **2** flip-flops. Only codes 00, 01, 10 are used. Code 11 never occurs, so its next state and output can be anything (X).
12	A Moore output depends only on the state, and the state changes only at the clock edge. Input noise between edges cannot change the state, so the output stays steady.

Next: Chapter 14 — Registers (SISO, SIPO, PISO, PIPO, bidirectional and universal shift registers).